Shared polynomial certificates for ZFC consistency

Round 18 research report and reproducible artifacts

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Abstract

This report gives new explicit polynomial presentations of syntactic ZFC refutation, and hence consistency via universal nonvanishing, with the logical calibration explicitly inherited. Spherical low-digit bounds and an odd-square radix produce quartics in 258 unrestricted integers, with sparse companions in 263, 285 and 286 integers, all of coefficient height 128446. Weighted natural digit pools yield quartics in 221–224 nonnegative integers, with dimension, support and height tradeoffs disclosed. Certified coefficient normalization, row-sign balancing and direct pairing of opposite monomial occurrences give height-one quartics in 571, 593, 646 and 742 unrestricted integers. Finally, fixed even radical weights and integral polynomial centers give eleven-integer equations at exact degree and binary-operation pairs 316/133, 308/140, 304/144 and 300/148. Complete new witness arguments, literal equations, authentic coefficient payloads and independent exact checks are supplied. The results are compared with six authenticated incoming packages at a fixed ProveIt snapshot. No proof of consistency, global minimality, or new end-to-end formal-kernel calibration is claimed.

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1 Results and interpretation

The new reductions use spare capacity already present in explicit arithmetic certificates. The signed construction pools low-digit bounds into spheres; the natural construction pools adjacent digit levels and can absorb the exact output bound. The height-one construction connects opposite monomial occurrences directly between rows and certifies fixed row-sign choices. The scalar construction changes a free integer coordinate by an integral polynomial shear. These are structural zero-set arguments, not assertions that a heuristic search has found a global optimum.

All numbers below refer to complete emitted artifacts on their stated domains. The direct and height-one polynomials have exact degree four. The scalar degree and live binary graph cost are separate measures.

1.1 Natural quartics

Endpoint	Variables	Rows	Terms	Height
`natural221`	221	171	13935	118464197
`natural222`	222	172	9917	83954634
`natural223`	223	173	7202	83954472
`natural224_sparse`	224	174	6527	83953869

All variables range over $\mathbb{N}$. The first three endpoints use four pools; the last uses three pools with the exact output bound absorbed. A certified row-order change supplies the 221-variable point. It is a genuine dimension/support/height tradeoff, not a simultaneous improvement over the balanced 222-variable form. The sparse 224-variable form improves variables, rows, support and height relative to the Round 17 sparse 230-variable point. The 222-variable form improves all four measures relative to the incoming Graph digits 226-variable point, while its 9917 terms exceed the Round 17 principal's 9184 terms. These distinctions matter when choosing an equation to use.

1.2 Unrestricted quartics

Endpoint	Variables	Rows	Terms	Height
Dimension	258	186	5364	128446
Grouped	263	191	4141	128446
Sparse merged	285	213	3750	128446
Sparse four pools	286	212	3744	128446

Every coordinate ranges over $\mathbb{Z}$. All four selected endpoints use the odd-square radix. The first three merge the t and $t+1$ spheres using two explicit product wires. The last keeps four spheres and thereby trades one additional variable for one fewer row and six fewer monomials. The 258-, 263-, and 285-variable points each improve variables, rows and support over their respective Round 17 282-, 287-, and 309-variable parents at unchanged coefficient height. These improvements do not require a new refutation calibration.

1.3 Height one quartics

Endpoint	Variables	Terms	Height	Constant
`compact571`	571	19833	1	1
`balanced593`	593	12027	1	1
`sparse646`	646	8368	1	1
`sparser742`	742	7226	1	1

All variables are unrestricted integers. The first two graphs use the new 263-variable signed source; the last two use the 285-variable source. Every collected nonzero coefficient is $+1$ or -1 and every declared variable occurs. The 571-variable point improves both dimension and support over the Round 17 793- and 806-variable height-one points; the 646-variable point does the same relative to its 994-variable sparse point. The four new choices retain different dimension/support tradeoffs. These are coefficient-height results, not claims of smaller dimension than the direct signed quartics.

1.4 Eleven integer arithmetic graphs

Endpoint	Variables	Degree	Operations	Fixed weight
Lowest new cost	11	316	133	8
First correction	11	308	140	8
Second correction	11	304	144	8
Lowest new degree	11	300	148	32

All eleven coordinates range over $\mathbb{Z}$, with the unchanged six authentic source34 numerical ports. Fixed even weights and integral polynomial centers lower exact degree from the Round 17 sharp endpoint of degree 328 and 130 operations, at higher graph cost. No variable-count reduction is claimed here. The exact new degrees are established for the full output polynomials at the actual coefficient tuple, including a surviving highest-degree monomial, not merely by syntactic degree propagation. The incoming 107- and 108-operation high-degree forms remain genuine low-cost alternatives. Expanded scalar support and total coefficient height are not assigned values because they were not materialized.

1.5 What is and is not newly proved

Each new witness argument is unbounded and proves both directions of the stated zero-set correspondence. Literal coefficient arrays, complete arithmetic graphs, full fixed numerical payloads, exact restoration certificates and independent finite checks accompany the proofs. The consistency interpretation is obtained by composition with the named inherited calibrations described next. No new kernel-certified logical endpoint, no proof of Con(ZFC), and no global simplicity minimum is asserted.

2 Logical scope and literal polynomial definitions

2.1 What the consistency interpretation asserts

Throughout, $\mathbb{N} = \{0, 1, 2, \ldots\}$ and $\mathbb{Z}$ denotes the unrestricted integers. The domain is part of an equation, not a presentation detail. An equation over $\mathbb{N}$ is not thereby an equation over $\mathbb{Z}$, and an unrestricted construction requires its own sign argument or an explicit domain conversion.

Let $\mathrm{Ref}_{\mathrm{ZFC}}(r)$ be the fixed arithmetized predicate saying that r codes a finite syntactic ZFC refutation. For each selected endpoint P on the stated domain D, the claim is

$$\exists \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) = 0 \quad \Longleftrightarrow \quad \exists r \in \mathbb{N} \; \mathrm{Ref}_{\mathrm{ZFC}}(r). \tag{1}$$

The ordinary transformations proved here are composed with the identified inherited calibration. On that proof chain, formalization in ZFC yields

$$\mathrm{ZFC} \vdash [\forall \boldsymbol{x} \in D^n \; P(\boldsymbol{x}) \neq 0] \longleftrightarrow \mathrm{Con}(\mathrm{ZFC}). \tag{2}$$

This report does not assert that either sentence in (2) is true. In particular it does not prove consistency, decide solvability, establish a global minimum, or supply a new end-to-end proof-assistant kernel check or encoded object-language ZFC derivation.

For the direct families, the retained ordinary logical interfaces are

`KEndpoint.c222_iff_zf_refutation`
`ConstructibleChoice.zf_refutation_iff_zfc_refutation`.

The first relates the fixed numerical calculus to a finite ZF refutation; the second is the retained ZF-to-ZFC refutation equivalence. The arithmetic front implements that calculus using finite syntax constructors, reciprocal CRT carriers, factorial and power certificates, and range tests. The scalar family instead retains the separately identified source34 native/compiler and refutation calibration, followed by its signed-rank and interpolation reductions. These are two calibrated routes, not interchangeable coefficient tuples.

2.2 The modular structure of the proof

The proof chain has four logically different layers.

1. The inherited calibration identifies a fixed finite arithmetic relation with syntactic refutation. Its provenance and exact numerical payloads are part of the source closure.

2. Unbounded arithmetic arguments prove both witness directions for the certificate relations. These include Pell classification, CRT choices, factorial fixed points, integer signs, and rank or interpolation lemmas.

3. Exact polynomial transformations replace one finite presentation by another. Integral substitutions, restored monic coordinates, graph identities and coefficient collection establish the claimed zero-set correspondence.

4. Finite independent checkers verify the actual emitted arrays and graphs, not merely an abstract instance of a theorem.

A hash certifies file identity, not a theorem; an exhaustive coefficient check certifies an identity, not an unbounded Pell assertion. Each layer is used only for what it establishes. The new sections give complete arguments for this round's transformations; the retained source appendix supplies the inherited arithmetic and logical inputs at their stated boundary.

2.3 Explicit finite arrays define the direct equations

For a direct endpoint, the ordered `variables` array in `equation.json` fixes the independent coordinates $x_0, \dots, x_{n-1}$. A term encoded as $[c, [i_1, \dots, i_d]]$ denotes $c \prod_{j=1}^{d} x_{i_j}$; the empty index list denotes the constant c. Repeated indices are retained and give ordinary powers. The m residual rows $R_0, \dots, R_{m-1}$ have integer coefficients and degree at most two. The equation is exactly

$$P(\boldsymbol{x}) = \sum_{j=0}^{m-1} R_j(\boldsymbol{x})^2 = 0. \tag{3}$$

Every coefficient and every coordinate appears in a finite supplied array. The adjacent `expanded_polynomial.json` stores the fully collected polynomial, including its constant term. There is no factorial, variable exponent, division, binomial coefficient, bounded quantifier, or sequence symbol among its algebraic primitives. Such expressions occur only in the proof explaining its zeros.

For either domain $D \subseteq \mathbb{Z}$, equation (3) is equivalent to $R_j(\boldsymbol{x}) = 0$ for every j, because each square is nonnegative. Exact degree four is established by a collected nonzero quartic coefficient, not merely by the upper bound $\deg R_j \le 2$. All independent coordinates are counted; named arithmetic abbreviations are not new unknowns.

The height-one endpoints are defined directly by their collected coefficient arrays. Their graph certificates prove the displayed energy identity and its zero-set correspondence. The independent checker reconstructs the energy and compares every collected coefficient. The height claim therefore concerns the final polynomial, not only its generating rows.

2.4 Arithmetic graphs define the scalar equations

The scalar graph leaves are eleven independent unrestricted integer variables, ordinary integer constants, or six fixed authentic numerical ports. Each internal node is binary addition, subtraction, or multiplication. An acyclic graph with one designated output therefore denotes a unique element of an ordinary polynomial ring over $\mathbb{Z}$. Registers abbreviate polynomials; they do not introduce hidden existential variables. There is no division or radical gate.

The source34 tuple is the inherited `mixed-parity-unrounded-1064921` tuple at coefficient source commit

`2605669a532342f153b13eb6b4a0af4b54f5fa37`.

Its ports include $\beta = 2^{1064921} - 1$, cone_A $= 2^{1113}$, cone_B $= 2^{666104}$, and the complete literal K_0, M_C, M_F payloads. The retained M_F already includes its native-value-plus-β shift. It must not be shifted twice or replaced by a port from a positive-domain tuple. The payload files contain complete minimal unsigned little-endian integers and are bound by byte length, bit length and SHA-256; neither a hash nor an ellipsis is substituted for a coefficient. The finite inherited compiler recipe reconstructs those values.

2.5 What each measure counts

For a nonzero collected polynomial $P = \sum_\alpha c_\alpha x^\alpha$, write

$$\deg P = \max_{c_\alpha \neq 0} |\alpha|, \qquad |\operatorname{supp} P| = \#\{\alpha : c_\alpha \neq 0\}, \qquad \operatorname{ht}(P) = \max_\alpha |c_\alpha|.$$

Support includes a nonzero constant coefficient. The number of residual rows is distinct from support and from the number of operations. Scalar cost counts live binary graph gates, including squares and multiplication by fixed constants; copying a register and reading a leaf are free. Dead nodes do not count. Unless an expanded scalar polynomial is materialized, its support and total coefficient height are unmeasured, not zero.

A point dominates another only when it is no worse in every displayed measure and strictly better in at least one, with domain and degree held fixed where required. This is a finite comparison among specified presentations. Height one, smaller dimension, sparse support and low graph cost are different objectives, and no single global simplicity ordering is claimed.

3 Spherical digit bounds over unrestricted integers

3.1 The exact inherited source and sign boundary

All coordinates in this section range over $\mathbb{Z}$. The source equations are the authenticated Round 17 centered dimension, grouped and sparse endpoints. Their literal residual arrays and integral coordinate translations are retained in `source_inputs`, bound by `SOURCE_PINS.json`. The new changes are spherical batching of the eleven digit bounds, an optional two-wire merger of adjacent denominator levels, and an optional odd-square radix replacing one initial sign guard. All other rows remain unchanged.

The inherited proof chain is `STRICT_BUDGET_PROOF.md`, `ORIENTATION_FIRST_PROOF.md`, `PROOF_315.md`, and the retained Round 16 arithmetic-front proof, followed by the named C222/ZF/ZFC calibration. In particular, the preceding signed frame and semantic anchor establish the canonical orientation of both coordinates of every principal Pell pair before using any of the eleven digit guards. This orientation-first fact is indispensable; the present construction is not an unrestricted reinterpretation of a natural system.

Let t, F, V, B, x, y, N, q denote the semantic values before the integral centering translations. The unchanged front has

$$t = 16 + h_0^2, \qquad V = (F^2+1)^t, \tag{4}$$

$$FB = V + s_0, \qquad B = r_0 + s_0 + 1, \tag{5}$$

$$y = V(V-1), \qquad x = F + 1 + (y+1)u_0, \tag{6}$$

$$x = 1 + [1 + (t+1)y]u_t, \qquad H_0 = 1 + \sum_{i=1}^{9} a_i, \tag{7}$$

$$N = B^x, \qquad q = (BH_0)^x. \tag{8}$$

Here the exponentiation equations indicate the interpretation of unchanged polynomial certificates, not primitives of the final polynomial. The factorial remainder and slack r_0, s_0 have retained nonnegativity guards; the other symbols initially have only their declared signed coordinate domains.

The square frame is $W = 2 + \sum(\text{power atom})^2 \geq 258$ and yields $|A| > W^W$. All requested signed exponents have absolute value below W. The squared semantic anchor bounds every classified Pell index. The orientation pool, its grouped form, or its separated sparse residues then recover $X_k(A), Y_k(A)$, allowing signed k and signed A. None of these arguments uses the eleven interval guards or the radix output-slack guard changed here.

Consequently the inherited signed power lemma is available: for a base $n \geq 2$, a framed signed index k and proposed integer output m, the exact power certificate forces $k \geq 0$ and $m = n^k$. If $k < 0$, multiplying the Pell congruence by n^{-k} and using the large modulus bound would force $mn^{-k} = 1$, impossible over the integers. For base one and known $k \geq 0$, the request gives $m = 1$. No output sign is presupposed.

In the baseline spherical variants, the old initial guard $q - B - 1 \geq 0$ is retained. The inherited pre-factorial schedule therefore gives

$$F \geq 1, \quad B \geq 2, \quad H_0 \geq 1, \quad y > F + 2, \quad x \geq 2 + (t+1)y, \quad N = B^x, \quad q = (BH_0)^x. \tag{9}$$

In particular $x \geq 6$ and $N \geq 16$. The odd-square variant obtains the same conclusions by a separate acyclic schedule below.

3.2 The old digit equations and their spherical replacement

Write $D_k = q^k$. The exact factorial-binomial remainder satisfies

$$(q+1)^{F^2+t} = (B + qH)D_t + r_f, \tag{10}$$

where $H \in \mathbb{Z}$ is a high quotient. Each of the ten range remainders has

$$(q+1)^a = ND_kQ_a + r_a, \qquad Q_a \in \mathbb{Z}. \tag{11}$$

The groups are

Denominator	Remainders	Number
D_t	factorial remainder and encoded index	2
D_{t+1}	encoded a, b, c, d and derived label h	5
D_{y+1}	the two stack representatives	2
D_x	the two strict quotient representatives	2

The old quadratic interval guards are

$$r(D - 1 - r) = a^2 + b^2 + c(c+1). \tag{12}$$

For $D > 0$, their right-hand sides are nonnegative, so they force $0 \le r < D$. Their exact source labels and centered coordinate identities are checked in the finite transformation certificate.

For a group of remainders with common denominator D, replace all its old guards by the single quadratic row

$$\boxed{D^2 - 2\sum_i r_i^2 = u^2 + v^2 + w(w+1).} \tag{13}$$

All three new coordinates are unrestricted. The right-hand side is nonnegative at every integer tuple. It also represents every nonnegative integer: by the three-square theorem, $4n+1$ is a sum of three squares; modulo four exactly one is odd. Writing the odd square as $(2w+1)^2$ and the other two as $(2u)^2, (2v)^2$ gives $n = u^2 + v^2 + w(w+1)$.

Equation (13) gives, for each member independently,

$$|r_i| \le D/\sqrt{2} < 3D/4. \tag{14}$$

It does not assume or immediately imply that r_i is nonnegative. The next argument supplies that conclusion using the exact digit equations.

3.3 Signed uniqueness of a small low tail

Lemma 3.1. *Let $q \ge 2$, $k \ge 1$, $D = q^k$, $a \ge 0$, and*

$$T(a,k) = \sum_{j=0}^{k-1} \binom{a}{j} q^j.$$

If $T(a,k) < D/8$, $r \equiv T(a,k) \pmod D$, and $|r| < 3D/4$, then $r = T(a,k)$.

Proof. The integer $r - T(a,k)$ is a multiple of D and satisfies

$$|r - T(a,k)| \le |r| + T(a,k) < 7D/8 < D.$$

It must be zero. □

Binomial expansion gives $(q+1)^a \equiv T(a,k) \pmod D$. Thus the literal rows (10) and (11) provide the required congruence independently of any sign on their quotients. The lemma explicitly excludes negative noncanonical representatives.

3.4 Recovering the factorial digit without circularity

For $M \ge 1$, $t \ge 0$ and $0 \le j \le t$,

$$\binom{M+t}{j} \le (M+1)^t. \tag{15}$$

For $j = t$, the product formula bounds each factor $(M+i)/i$ by $M+1$. For $j < t$, induction on t and Pascal's identity bound the two summands by $2(M+1)^{t-1} \le (M+1)^t$; the boundary cases are immediate.

The unchanged t request at base $F^2 + 1 \ge 1$ first yields $V = (F^2+1)^t \ge 1$. The retained ceiling relations and $B \ge 1$ force $F \ge 1$. Since $FB \ge V \ge F^2 + 1$, we have $B > F$, hence

$$V \le FB < B^2.$$

The endpoint gives $x \ge 6$, and the exact radix gives $q \ge B^x$. Therefore

$$B^2 \le q/16, \qquad V < q/16. \tag{16}$$

Applying (15) with $M = F^2$, every coefficient through position t is at most V. Consequently

$$T(F^2 + t, t) < \frac{VD_t}{q-1} < \frac{qD_t}{16(q-1)} \leq D_t/8. \tag{17}$$

No factorial identity has been assumed. The D_t sphere bounds r_f even while its other coordinate remains uninterpreted, so Lemma 3.1 gives $r_f = T(F^2 + t, t)$. Subtract this tail from (10) and reduce modulo qD_t to obtain

$$B \equiv \binom{F^2 + t}{t} \pmod{q}.$$

The old output guard gives $0 \leq B < q$ in the baseline construction; the odd-square schedule below proves $B < q$ directly. The binomial lies in $[0, q)$ by (16). Thus $B = \binom{F^2+t}{t}$. Only now is the inherited square-radius factorial fixed-point theorem applied to the unchanged ceiling inequalities, yielding $F = t!$ and the exact reciprocal-CRT product B.

3.5 Recovering the other ten intervals in order

First handle ordinary encoded representatives, including the encoded-index representative C_0. They are already nonnegative, and $H_0 \geq 1$. For every ordinary representative a and requested position $0 \leq j \leq k \leq x$,

$$\binom{a}{j} \leq \max(1, a)^k \leq H_0^x = q/N.$$

All four lower indices are at most x, by $x \geq 2 + (t+1)y$ and $y \geq 1$. Since $N \geq 16$,

$$T(a, k) < \frac{qD_k}{N(q-1)} \leq D_k/8. \tag{18}$$

The spheres and Lemma 3.1 recover these ordinary tails, in particular the tail for C_0. Independently, the retained literal reciprocal-index congruence row gives $B \mid C_0 + F^2 + 1$. Since the now exact factorial product satisfies $B > F^2 + 1$, $C_0 = 0$ is impossible. Thus $C_0 \geq 1$ and $H_0 \geq 2$. The raw congruence does not itself depend on a remainder interval; stating the ordinary tails first also makes clear that none uses the stronger derived-label majorant.

For the derived label, the unchanged constructor ledger gives

$$h \leq H_0^\ell, \qquad \ell = 4194304.$$

The inherited factorial estimate $\ell(t+1) < 2F$ and endpoint bound $2F < x$ imply $\ell(t+1) < x$. Its needed binomial coefficients through position $t+1$ are therefore also at most $H_0^x = q/N$. The same strict tail estimate identifies its remainder. All eleven old interval conclusions have now been recovered.

More explicitly, after subtracting a range tail, divide (11) by D_k. The expansion becomes

$$NQ_a = \binom{a}{k} + qH_a$$

for an integer H_a. Since $N \mid q$, this yields $N \mid \binom{a}{k}$, exactly the inherited range predicate. The unchanged prime-power precision, residue extraction and local Horn flow arguments then yield the original finite-refutation interpretation. No signed quotient has been treated as natural before this recovery.

3.6 Both witness directions and adjacent-level merging

For completeness, start with a finite refutation and construct the inherited cofinal semantic witnesses. Choose all digit remainders canonically as $T(a,k)$. By (17) and (18), each is less than its denominator divided by eight. In a group with $m \le 5$,

$$2\sum_i T_i^2 < \frac{m}{32}D^2 < D^2.$$

The integer $D^2 - 2\sum_i T_i^2$ is nonnegative, so its ternary representation supplies the new guard coordinates. All unchanged residuals already vanish. Conversely, the preceding soundness proof restores every old interval from a new zero. Each $r(D-1-r) \ge 0$ then has a ternary representation, filling the deleted old coordinates afresh. Undoing the recorded integral translation gives an actual zero of the calibrated source. This proves integer existential equivalence, without claiming that arbitrary old private witnesses extend unchanged.

For the optional merger, introduce two explicit product coordinates

$$z_f = qr_f, \qquad z_C = qr_C.$$

Replace the first two spheres by one at denominator $D_{t+1} = qD_t$, using z_f, z_C and the five old D_{t+1} remainders. The new sphere gives $|qr_f|, |qr_C| < 3qD_t/4$, hence the old radius bound because $q > 0$. The soundness argument is unchanged. Canonical scaled tails are below $D_{t+1}/8$, and now

$$2\sum_{i=1}^{7} T_i^2 < \frac{7}{32}D_{t+1}^2 < D_{t+1}^2,$$

so completeness still holds. Two ternary triples become one, saving three coordinates, while two product wires are added: net one fewer coordinate and one additional residual. Every residual remains quadratic.

3.7 An odd square replaces the initial radix sign guard

The separately identified odd-radix variants remove the three-coordinate ternary guard for $q - B - 1 \ge 0$ and add one unrestricted z with

$$q = (2z+1)^2. \tag{19}$$

This supplies only $q \ge 1$ initially. The following schedule replaces the old use of $q \ge 2$.

1. The unchanged frame, anchor and orientation proof gives canonical principal Pell pairs without the removed guard. Since $t \ge 16$, the request at base F^2+1 yields $V = (F^2+1)^t$, including the base-one case. The retained ceiling guards then give $F \ge 1$, $B \ge 2$, $V < B^2$ and $y > F+2$.

2. Now $q+1 \ge 2$. Each actual encoded representative is the exponent of a numerator request at this base, so the signed power lemma makes every such representative nonnegative. Hence $H_0 \ge 1$. The derived-label request similarly gives a nonnegative index, with zero already excluded by the squared anchor.

3. Since $BH_0 \ge 2$, its existing x request forces $x \ge 0$ and $q = (BH_0)^x$.

4. In $x = F+1+(y+1)u_0$, a value $u_0 \le -1$ would imply $x \le F - y < 0$. Thus $x \ge F+1 \ge 2$. The other endpoint forces $u_t \ge 1$, so $x \ge 2 + (t+1)y$. Consequently $q \ge B^x \ge B^2 > B$ and $q \ge 4$.

5. The remaining powers, including $N = B^x$ and every denominator, are now exact. The missing output slack $q - B - 1$ is nonnegative before factorial decoding. The preceding spherical proof applies in its stated order.

No removed guard is used to establish its own replacement.

For completeness, choose the beta code x even in the inherited cofinal CRT construction. Since $y = V(V-1)$ is even, every beta modulus $1 + (i+1)y$ is odd, and adding their odd common CRT modulus changes parity. A sufficiently large even representative preserves all encoded values and nonnegative endpoint quotients. After $F = t!$, each factor of B is odd, so N^2 is odd. Every later encoded representative can be changed by N^2 while preserving its required congruences. Choose a, b of opposite parity, as already needed for even semantic-anchor index h. Independently choose one of the other seven parities so the sum of all nine representatives is even; then H_0 is odd.

The dominant a can still lie in $[2N^2, 4N^2)$ with its prescribed parity, and the other representatives remain at most $2N^2$. Thus the inherited anchor dominance and cap estimates remain valid. Now BH_0 is odd and x even, so the positive integer $R = (BH_0)^{x/2}$ is odd. Taking $z = (R-1)/2$ gives (19). All other CRT, radix and Pell witness choices proceed unchanged. This proves that the square restriction is compatible with completeness, not merely sufficient for soundness.

3.8 What the finite checker establishes

The producer reads only copied, hash-bound source equations and their coordinate translations. The separately implemented checker imports no producer. It reconstructs each replacement in named coordinates, verifies every untouched residual, expands each square and collects all coefficients, and checks liveness, degree, support and height. Deliberate changes to sphere, frame and odd-square rows must be rejected. Finite Pascal or tail examples serve as diagnostics only; the preceding unbounded arguments prove the mathematical implication. Together with the retained ordinary calibration, this establishes (1) for each selected unrestricted quartic.

4 Pooled digits in the aligned self sized radix

4.1 Exact inherited interface and the changed rows

All coordinates in this section range over $\mathbb{N}$. The input is the Round 17 normalized raw quadratic graph for `strict_natural229`, with 334 variables and 284 rows. Its complete inherited ordinary proof is `ONE_ROW_SEMANTIC_ANCHOR_PROOF.md`. The present transformations alter only the eleven private modular remainder bounds and, in one variant, the exact binomial output bound. The six exact power requests, semantic Pell anchor at the derived label J, and simultaneous index certificate are unchanged.

Use the inherited notation

$$t \geq 12, \quad V_0 = (F^2+1)^t, \quad K = x, \quad N = B^K, \quad q = (NS)^K.$$

Here S is one plus the sum of the nine positive independent CRT representatives, not an individual stack representative. In genuine completed witnesses,

$$F = t!, \qquad B = \binom{F^2+t}{t}.$$

These two identities are not assumed prematurely in soundness. Write

$$D_0 = q^t, \quad D_1 = q^{t+1} = qD_0, \quad D_y = q^{y+1}, \quad D_x = q^x.$$

There are eleven modular remainders, organized as follows.

Remainders	Common denominator	Number
r_0, r_C for the factorial digit and index column	D_0	2
r_a, r_b, r_c, r_d, r_J	D_1	5
r_{S_0}, r_{S_1} for the two stack columns	D_y	2
r_u, r_v for quotient columns	D_x	2

The old separate bounds are $D_i = r_i + s_i + 1$ with $s_i \in \mathbb{N}$. The exact factorial digit additionally satisfies $q = B + o + 1$, with $o \in \mathbb{N}$.

4.2 A coefficient budget lemma

Lemma 4.1. *Let $q \geq 2$, $k \geq 1$, and*

$$r_i = \sum_{j=0}^{k-1} c_{ij} q^j, \qquad c_{ij} \in \mathbb{N}.$$

If $\sum_i c_{ij} \leq q - 1$ for every j, then $\sum_i r_i \leq q^k - 1$. Hence

$$q^k = \sum_i r_i + s + 1 \tag{20}$$

has a natural slack s. Conversely, for arbitrary natural r_i, s, (20) implies $r_i < q^k$ for every i, and the old separate slacks are restored by the nonnegative polynomial formulas

$$s_i = s + \sum_{h \neq i} r_h.$$

Proof. Summing by positions gives

$$\sum_i r_i = \sum_{j=0}^{k-1} \Big(\sum_i c_{ij} \Big) q^j \leq (q-1) \sum_{j=0}^{k-1} q^j = q^k - 1.$$

This proves existence of $s = q^k - 1 - \sum_i r_i \geq 0$. Conversely, each r_i is a nonnegative summand of $q^k - s - 1 < q^k$. The displayed restoration satisfies $r_i + s_i + 1 = q^k$ identically. □

The completeness condition bounds the sum of the coefficients, not merely each remainder separately. Separate old inequalities alone would not justify pooling.

4.3 Uniform spare capacity in the actual radix

Construct genuine witnesses in the inherited noncircular order. Begin with a finite derivation and choose its padded horizon $t \geq 12$. Fix $F = t!$, B, $K = x$, and $N = B^K$; choose the nine positive independent CRT representatives modulo N^2. A further lift of the dominant a representative by N^2 preserves every required congruence and ensures $S > N^2$. The inherited cofinal choice is retained, so the semantic anchor still dominates the required indices. Every independent representative is strictly less than S. The unchanged endpoint and ceiling arguments give

$$K = x \geq F + 1 \geq 2, \qquad B > F \geq 1, \qquad N \geq 4.$$

Consequently

$$q = N^K S^K \geq 16 S^K. \tag{21}$$

For an ordinary representative $n < S$ and a requested lower digit position $0 \leq j \leq k \leq K$,

$$\binom{n}{j} \leq \max(1, n^K) < S^K.$$

The same assertion for the derived label uses the inherited positive arithmetic-graph majorant

$$J \leq S^\ell, \qquad \ell = 2^{22}, \qquad \ell(t+1) < F < K.$$

For $0 \le j \le t+1$, this gives $\binom{J}{j} \le \max(1, J^j) < S^K$. For the factorial upper index, the inherited elementary estimate yields

$$\binom{F^2+t}{j} \le V_0 \quad (0 \le j \le t), \qquad V_0 < B^2 \le N.$$

Since $S > N^2$, both V_0 and B are strictly below S^K. Thus every low-position coefficient used in the eleven remainders, and also the exact output B, is at most $S^K - 1$.

These are completeness estimates, after genuine witnesses are fixed. The unchanged common-root Pell congruence identifies $(q+1)^n$ modulo the requisite digit modulus. Since the relevant binomial coefficients are below q, the canonical remainders are exactly the low base-q polynomials used above. This explanation introduces no exponentiation operation into the emitted quadratic rows.

4.4 Four independent pools

Replace each group's old private bounds by one row

$$D_G = \sum_{i \in G} r_i + s_G + 1, \qquad s_G \in \mathbb{N}. \tag{22}$$

The largest group has five remainders, so at every position the coefficient sum is at most $5(S^K - 1) < q$. Lemma 4.1 supplies the four new slacks for canonical witnesses.

For soundness, apply the converse part of the lemma independently to each pool. Every deleted old slack is restored by $s_i = s_G + \sum_{h \in G \setminus \{i\}} r_h \ge 0$. No binomial interpretation is needed in this direction. All other rows and coordinates are unchanged, so a zero of the pooled raw graph yields a zero of the exact inherited graph. For completeness, construct canonical inherited witnesses first and then use the coefficient bounds above. The raw graph loses seven variables and seven rows: its size becomes 327 variables and 277 rows.

4.5 Weighted mixing of adjacent denominator levels

The identity $D_1 = qD_0$ lets the first two groups share a single bound. Put $T = r_a + r_b + r_c + r_d + r_J$. Replace their seven old bounds by

$$qD_0 = q(r_0 + r_C) + T + s + 1, \qquad s \in \mathbb{N}. \tag{23}$$

Keep the D_y and D_x pools from (22). For soundness, $q \ge 1$ and $D_0 \ge 1$ imply

$$qr_0 < qD_0, \quad qr_C < qD_0, \quad r_a, r_b, r_c, r_d, r_J < D_1.$$

Thus $r_0, r_C < D_0$, and all seven old slacks $D_i - r_i - 1$ are nonnegative integers. Their substitution restores the old equations identically.

For completeness, multiplication by q shifts each of the two smaller remainders' coefficient arrays upward by one position. All terms lie below position $t+1$. At most seven original coefficients contribute at any position, so their sum is at most $7(S^K - 1) < q$. Lemma 4.1 provides s. This raw graph has 326 variables and 276 rows.

The required positivity is available before decoding any binomial. The retained frame and anchor first identify all six exact powers. The unchanged ceiling equations imply $B \ge 1$, $V_0 \ge 1$, $F \ge 1$, $B > F$, and $K \ge 2$. Hence $N = B^K \ge 4$ and $q = (NS)^K > 0$. This is precisely the inherited bootstrap; it uses neither a removed remainder bound nor the separate exact digit output bound.

4.6 Absorbing the exact binomial output bound

A further compatible change replaces (23) by

$$qD_0 = q(r_0 + r_C) + T + BD_0 + s + 1 \tag{24}$$

and deletes $q = B + o + 1$ and its private o. In canonical witnesses, BD_0 contributes the single digit B at position t. At each position, at most eight quantities bounded by $S^K - 1$ contribute. Therefore their sum is less than $8S^K \le q/2 < q$, proving completeness by the same lemma.

For soundness, the nonnegative extra term BD_0 cannot weaken the remainder inequalities already proved. The new row also gives $BD_0 < qD_0$. Since $D_0 > 0$, it follows that $B < q$. Thus $o = q - B - 1 \in \mathbb{N}$ restores the deleted exact output bound. The pre-decoding positivity argument in the preceding subsection remains unchanged. This raw graph has 325 variables and 275 rows. All displayed rows remain quadratic: qr_i and BD_0 are products of existing coordinates, not additional quantified gates.

4.7 Exact projection to the emitted quartics

Each selected endpoint is obtained from one of these raw graphs by a finite sequence of monic polynomial eliminations. Every restored coordinate is certified as a positive rational combination of squares of affine rational forms plus a polynomial with nonnegative integer coefficients. Such a formula is nonnegative on $\mathbb{N}^n$. Its exact equality to the integer restoration polynomial is checked after expansion, so integrality follows from the restoration polynomial itself.

The independent checker replays all substitutions and retained rows, checks each nonnegativity certificate, verifies that residual degree never exceeds two, and collects the entire final sum of squares. A changed elimination order is supplied as an explicit permutation of the raw rows; its permutation property and exact relation to the fixed pooled source are checked. No heuristic search result is used as a substitute for the restoration identities.

Composing the raw witness constructions with these certified projections proves existence equivalence with the inherited natural graph and therefore (1). Arbitrary old zeros need not extend to pooled zeros with the same private remainders; the completeness argument constructs canonical witnesses afresh from the encoded finite refutation. This is existential equivalence, not an asserted bijection of every intermediate zero set.

4.8 Attribution and the precise new composition

The incoming `conzfc_graph_digits_2026-10-24` package supplies the four-pool digit-budget idea for an older semantic cap. Round 17 supplies the aligned $K = x$ radix, factorial/product fusion, strict quotient ranges, cofinal representative choice and literal parent graph. The new composition uses the stronger margin (21), weighted mixing of adjacent denominator levels, and absorption of the exact output bound. The retained contractor and independently implemented projection checker are credited to the earlier package100 code. None of these attributions alters the inherited logical calibration.

5 Height one quartics by signed occurrence fusion

The new reduction acts on a finite quadratic system over unrestricted integers. It changes neither its integer zero set up to a specified bijection nor its inherited consistency calibration. Its new ingredients are row-sign balancing and *opposite-occurrence fusion*: equal monomials carrying opposite source signs are represented by a single incidence edge between their source rows. The retained Round 17 incidence compiler routed both occurrences separately to a common hub; the incoming Graph digits package also used occurrence pairing during alias identification. Here the pair itself becomes a direct source-row edge. All claims about coefficient height below refer to complete collected coefficient files, not merely an arithmetic circuit.

5.1 Certified normalization and row orientations

Let the given source be $R_r(x) = 0$. We first use the preceding exact affine refactorization, when applicable: if the homogeneous quadratic part of a row is $\eta L(x)^2$, with $\eta = \pm 1$, introduce $z = L(x) + k$ and replace the row by $\eta z^2 + R_r(x) - \eta(L(x) + k)^2 = 0$. The new definition is monic and its value is uniquely determined. Shared affine definitions can be reused. Coefficient-by-coefficient substitution verifies every transformed row. The provided original source, transformed source and factor certificate fix this operation completely.

Normalize integer coefficients by monic definitions of integer constants $t_c = c$ and scale variables $s_{c,i} = cX_i$. A constant is defined from already defined smaller constants as $t_c = t_a + t_b$ when $a + b = c$, using $t_1 = 1$. If no such pair is available, introduce the smaller binary intermediate and use $t_c = t_{\lfloor c/2 \rfloor} + t_{\lfloor c/2 \rfloor} + (c \bmod 2)$. Each added definition is triangular and monic. The delivered DAG prefers two already-needed constants; in particular it does not introduce a binary intermediate merely because a binary-only recipe would do so.

For each absolute quadratic coefficient $c > 1$, form a finite graph on the source coordinates, with an edge $\{i, j\}$ for every term cX_iX_j. A loop at i requires selecting i. Choose any vertex cover V_c of this graph, and introduce $s_{c,i} = t_cX_i$ only for $i \in V_c$. A term cX_iX_j is then represented by $s_{c,i}X_j$ if i was selected, or by $X_is_{c,j}$ otherwise. The cover condition proves that one option always exists. The supplied deterministic greedy cover starts with the loop vertices and repeatedly selects a highest-degree remaining vertex; optimality is not assumed. Constants and linear terms use t_c and t_cX_i, respectively. Unit coefficients need no constructor.

Splitting repeated summands into separate occurrences yields

$$S_r(y) = \sum_j \epsilon_{rj} m_{rj}(y), \qquad \epsilon_{rj} \in \{-1, 1\}, \quad \deg m_{rj} \le 2.$$

Every normalized coordinate has a unique integral value over an original tuple. The checker verifies all triangular pivots and each exact source-row pullback, rather than trusting the constant-chain or cover-selection recipe. Thus this refinement saves constructors without changing either witness map. The coefficient-factor choice, just like the later graph, is fixed by the certificate and is not a parameter of the polynomial.

Choose a fixed sign $\sigma_r \in \{-1, 1\}$ for each normalized row, and replace S_r by $T_r = \sigma_r S_r$. This preserves both its zero set and uniqueness of all constructors. The choice is part of the finite certificate, not a parameter of the final polynomial. The supplied heuristic flips a row when doing so reduces

$$\sum_m \left| \sum_{r,j:m_{rj}=m} \sigma_r \epsilon_{rj} \right|.$$

This balances opposite occurrences of the same logical monomial. Correctness does not require optimality of this heuristic.

5.2 Direct edges for cancelling occurrences

Create disjoint vertex blocks B_r for the rows and a hub block B_0. Inside each block choose a tree, with one independent unrestricted integer flow variable on each tree edge. A normalized occurrence is assigned to a vertex in its own row block. Partition the occurrences into sets of size one or two. A pair is allowed only when its occurrences are in distinct rows, have the same logical monomial, and have opposite signs in T_r.

A singleton of sign ϵ is represented by an edge from its row vertex to a hub vertex carrying ϵM. A pair $+m$ in row r and $-m$ in row s is represented by one edge from its row-r vertex to its row-s vertex, carrying M. Here M is a degree-two homogeneous monomial in aliases of the logical factors of m: a linear term is multiplied by a unit alias and a constant uses two unit aliases. Each factor alias has one declared logical value. The two occurrences of a pair use *the same* edge and the same factor aliases.

The graph is required to be simple. The constructor rejects a pairing if its proposed unordered vertex pair was already used; remaining hub edges are placed without parallel edges. The certificate lists every bound source occurrence, and the checker verifies that each appears exactly once. More precisely, the edge boundary at the row-block level equals its list of signed source occurrences. Consequently, after substituting the intended logical values,

$$\sum_{v\in B_r} b_v = T_r(y), \qquad \sum_{v\in B_0} b_v = -\sum_r T_r(y). \tag{25}$$

where b_v is the divergence from prescribed edges alone. A fused edge contributes $+M$ and $-M$ to precisely its two bound rows; hence (25) is unchanged by fusion. This is the central new identity.

5.3 The coefficient-height-one energy

Let d_v denote the outgoing-minus-incoming sum of all prescribed and free edges, and put

$$E = \frac{1}{2}\sum_v d_v^2.$$

Each edge has two endpoints, so its diagonal energy coefficient is one. Two distinct edges of a simple graph meet at most once, so their formal cross coefficient is 0 or ± 1. Give each prescribed product an initial private factor alias, and share its other logical factor as in the previous incidence compiler. The initial fully expanded energy has height one. Subsequently identify only aliases of the same logical value, and accept an identification only when full exact coefficient collection still has coefficients in $\{-1, 0, 1\}$. Such substitution preserves the sum-of-squares identity even when signed collisions cancel.

For every remaining non-unit logical class with at least three aliases use

$$C(a_1, \dots, a_k) = \sum_i a_i^2 - \sum_i a_i a_{i+1}, \quad a_{k+1} = a_1.$$

Since $2C = \sum_i (a_i - a_{i+1})^2$, it is nonnegative and vanishes exactly when all aliases agree. A singleton needs no penalty. A class with two aliases is either completed to a three-cycle by one dummy, or uses

$$Q(a, b) = a^4 + b^4 - a^3 b - ab^3 = (a - b)^2(a^2 + ab + b^2).$$

This is nonnegative and vanishes exactly at $a = b$; also $2Q = (a^2 - b^2)^2 + (a^2 - ab)^2 + (ab - b^2)^2$. The latter option is used only when neither alias occurs as a prescribed square, which ensures its four monomials cannot collide with the incidence energy. For all unit aliases, adding dummies if necessary so that $\ell \geq 2$, put

$$U = 1 + \sum_{i=1}^{\ell} u_i^2 - u_1 - u_\ell - \sum_{i=1}^{\ell-1} u_i u_{i+1}.$$

Twice U is $(u_1 - 1)^2 + \sum_{i<\ell}(u_i - u_{i+1})^2 + (u_\ell - 1)^2$. Thus the final literal polynomial is

$$\boxed{P = E + \sum C + \sum Q + U.} \tag{26}$$

It is nonnegative over the reals. All monomials have total degree at most four. The full coefficient files are independently checked to have exact degree four, every nonzero coefficient equal to $+1$ or -1, constant coefficient one, and every stated variable present. The new fusion step changes the source-edge incidence graph, but requires no change to these equality penalties or their proof.

5.4 Forward map, inverse map, and uniqueness

Given a source tuple, evaluate the monic affine, constant and scale constructors, assign each ordinary alias its logical value and every unit alias the value one. Assign equality dummies the value of their class. All equality and unit penalties then vanish.

Root each internal tree at its first listed vertex. For a tree edge with child v, let $\tau_v = \pm 1$ be its incidence sign at the child, and set

$$w_v = -\tau_v \sum_{t \text{ in the subtree of } v} b_t.$$

This is an explicit integral polynomial in the source coordinates. Eliminating leaves proves that all nonroot divergences vanish. Formula (25) gives root divergence T_r in each row block and $-\sum_r T_r$ in the hub block. Thus the canonical lift satisfies, universally,

$$P(L(x)) = \frac{1}{2}\left(\sum_r R_r(x)^2 + \left(\sum_r \delta_r R_r(x)\right)^2\right). \tag{27}$$

where δ_r is the certified sign of the normalized row descending from the original residual R_r. Constructor rows pull back to zero. In particular every source zero has an integer lift. The affine-definition rows also vanish identically under their canonical substitution, so (27) composes unchanged through affine refactorization. One must not replace (27) by a bare sum of the original squares.

Conversely, if $P(z) = 0$, nonnegativity in (26) forces every equality penalty and U to vanish. Hence all ordinary aliases in a class agree, and all unit aliases equal one. Also $E = 0$ forces every vertex divergence to vanish. Summing them over a row block cancels its internal tree flows; by the certified occurrence-boundary identity the result is precisely $T_r(y) = 0$. Since $\sigma_r = \pm 1$, all normalized rows $S_r(y)$ vanish. Triangular monic constructors then force the constant, scale and affine coordinates to their intended values. Reading the original logical coordinates yields an original source zero.

The same argument proves uniqueness: constructors, aliases, units and dummies have unique values over the original tuple, and leaf elimination uniquely determines every internal tree flow. Thus the displayed forward map and coordinate projection are inverse bijections between integer zero sets. No sign restriction on a final variable, division by a variable, or unproved boundedness assertion is used.

5.5 Evidence and inherited consistency scope

The independent checker imports no producer. It verifies source hashes, all signed row pullbacks, topological constructors, every occurrence and its edge boundary, graph simplicity, every internal tree, equality-class coverage, all coefficients through literal half-sum-of-squares expansion, and every vertex of the universal symbolic lift. Its tests use explicit exceptions and remain active under optimized Python. The complete polynomial is also supplied in ordinary literal notation. Neither this finite algebra nor its successful replay proves Con(ZFC): it transfers the supplied source's existing ZFC-provable equivalence between unsolvability and consistency. No global-optimality, literature-wide priority or new end-to-end formal-kernel calibration is claimed.

6 Whole expression centers in eleven integers

6.1 Complete polynomial definitions

The independent coordinates are exactly

$$(j, C, z, h, r, v, w, T, c, n, s) \in \mathbb{Z}^{11}.$$

The six fixed ports are the authentic source34 numerals specified in Section 2; none is another variable. Define the following ordinary polynomial abbreviations:

$$\begin{aligned}
J &= 6j+1, & q &= \beta J+1, & W &= 12w+4,\\
X &= qW, & Y &= v, & E &= XY,\\
a &= E+Y, & A &= a+2, & H &= 4a+3,\\
\Delta &= a^2+H = A^2-1,
\end{aligned}$$

$$\begin{aligned}
F &= (K_0+W)C - z(q-1),\\
R &= [q(q-F)-C](q^2-1) + (M_C+qM_F)J,\\
k &= R+1+hE, \qquad \eta = c - Yk,\\
D_{\mathrm{err}} &= (\mathrm{cone_A}F-q)^2 + (\mathrm{cone_B}C-q)^2,\\
B_0 &= k^2(q^2-D_{\mathrm{err}}) - 4q^2\eta^2,\\
\mathcal{B} &= Y^2B_0 - k^2q^8, \qquad D = \mathcal{B}-4,\\
\ell &= XY^2, \qquad D_t = 1+\ell(\ell+1)k^2,\\
D_g &= 1+DD_ts^2, \qquad P = D_tD_g.
\end{aligned}$$

The cofactor, auxiliary norm and carrier registers are

$$\begin{aligned}
D_m &= 1+\Delta c^2, & B &= rAD_m, & U &= rc^2D_m,\\
I_2 &= c^4B = Ac^2U, & D_f &= 1+\Delta I_2, & Q &= \Delta^2 I_2,\\
\delta &= Q-1, & V &= D_f(c-R+4AUT)-c, & C_y &= QV^2-1,\\
K &= X(3c-2X)+2, & M &= 2q^3, & D_c &= HM,\\
S &= srHM^5,
\end{aligned}$$

$$z_c = D_cT + MK + H(v-q^3), \qquad A_c = (z_c^2+3D_c^2)(z_c^3-19D_c^3).$$

Choose one fixed even weight L and one center Ψ below:

$$\begin{aligned}
\Psi_{316} &= [(8\Delta+4)U - rc^4]V, & L_{316} &= 8,\\
\Psi_{308} &= \Psi_{316} + 2U^3T(c^2-2), & L_{308} &= 8,\\
\Psi_{304} &= \Psi_{316} + U^2T[2U(c^2-2)+16A^3], & L_{304} &= 8,\\
\Psi_{300} &= [(32\Delta+16)U - 4rc^4]V & & \\
&\quad + U^2T[8U(c^2-2)+64A^3+3rc^6], & L_{300} &= 32.
\end{aligned}$$

Every center is an integer polynomial independent of n. In each case put

$$\begin{aligned}
Z &= S(n+\Psi) - A_c,\\
p_a &= \delta Z^2, \qquad p_b = \delta S^2P, \qquad p_c = L^2S^2BD_fC_y,\\
\mathcal{N}_{L,\Psi} &= (p_a+p_b-p_c)^2 - 4p_ap_b.
\end{aligned} \tag{28}$$

The equation is $\mathcal{N}_{L,\Psi}=0$ over $\mathbb{Z}^{11}$. No division, radical or variable exponent is an operation of this polynomial. The literal graph shares the identities

$$P = D_t + (\mathcal{B}-4)(D_ts)^2,$$

$$\mathcal{B} = (Yk)^2(q^2-D_{\mathrm{err}}) - q^2[4(Y\eta)^2 + (kq^3)^2], \qquad A^2 = \Delta+1.$$

Sharing these registers changes graph cost, not the polynomial.

6.2 An even weight and integral center lemma

Theorem 6.1. *For every fixed nonzero even L and every integer polynomial Ψ in the ten coordinates other than n, the projection of the integer zero set of* (28) *onto those ten coordinates equals that of the retained Round 17 free-scale equation. The latter uses $L = 2$ and $\Psi = (2\Delta + 1)UV$.*

Proof. We first record the unconditional arithmetic facts from Sections 2–4 of the retained `LINEAR_COFACTOR_ROUND17_PROOF.md`. They do not assume a zero of the present norm. The authentic affine restrictions give

$$\begin{aligned} q &\equiv 0 \pmod 2, & q &\equiv 2 \pmod 3, & |q| &\geq 16, \\ X &\equiv 0 \pmod 8, & X &\equiv 2 \pmod 3, & |X| &\geq 4|q|. \end{aligned}$$

Thus $A \neq 0$, $\Delta = A^2 - 1 > 1$, $\Delta \equiv 0 \pmod 3$, H is odd and nonzero, and $D_m, D_t \geq 1$. The exact modulo-sixteen budget identity gives $D \equiv 8$ or $12 \pmod{16}$, so D is nonzero and nonsquare. Also $P \equiv 1 \pmod 4$ and $P \neq 0$. Both D_f and δ are nonzero, Q is even, and δ is odd.

The rational polynomial $\pi(t) = (t^2 + 3)(t^3 - 19)$ has no rational root. Since $D_c \neq 0$,

$$A_c = D_c^5 \pi(z_c/D_c) \neq 0.$$

If $S = 0$, the output norm is $(\delta A_c^2)^2 \neq 0$. Every new zero therefore has $S, s, r, B \neq 0$.

Put $D_a = \delta C_y$ and $C_* = BD_fD_a$. One has $D_a > 0$: if $c = 0$, then $Q = 0$ and $D_a = 1$; if $c \neq 0$ and $B < 0$, then $\delta, C_y < 0$; if $c \neq 0$ and $B > 0$, then $D_f > |c| + 1$, so $V = D_f b - c$ is nonzero for every integer b, and $\delta, C_y > 0$. Moreover $BD_f > 0$ whenever $c \neq 0$. Hence $C_* \neq 0$, and its only possible negative case is $c = 0, B < 0$.

The four-conjugate norm identity for (28), after normalizing by nonzero S and multiplying by δ, gives a rational signed sum

$$\pm L\sqrt{C_*} \ \pm \delta\sqrt{P} \in \mathbb{Q}. \tag{29}$$

If exactly one radicand is negative, its nonzero imaginary contribution cannot cancel. If both are negative, then $c = 0$, $\delta = -1$ and $C_* = B$. Cancellation would force $L^2B = P$, impossible modulo four because L is even and $P \equiv 1 \pmod 4$. Therefore $C_*, P > 0$.

Write these positive integers as squares times positive squarefree integers. Different nontrivial squarefree radicals are independent over $\mathbb{Q}$. The only remaining possible irrational cancellation would have the same nontrivial squarefree part in both terms of (29). Its first coefficient would be even, whereas its second coefficient would be odd: L is even and δ, P are odd. Cancellation is impossible. Thus C_* and P are positive integer squares.

The cofactor coprimalities are unchanged:

$$\gcd(B, D_f) = \gcd(B, D_a) = \gcd(D_f, D_a) = 1.$$

Indeed $D_f \equiv 1 \pmod B$ and $Q \equiv 0 \pmod B$, giving $D_a \equiv 1 \pmod B$. Modulo D_f, one has

$$Q \equiv -\Delta, \quad \delta \equiv -A^2, \quad V \equiv -c, \quad C_y \equiv -D_m, \quad D_a \equiv A^2 D_m.$$

Since $B = rAD_m$, while $D_f \equiv 1$ modulo both A and D_m, these imply the last coprimality. Therefore $|B|, |D_f|, D_a$ are squares. As $D_a > 0$, B, D_f have the same sign. But $D_f \equiv 1 \pmod 3$ excludes a negative square, so there are positive integers u, f, e with

$$B = u^2, \qquad D_f = f^2, \qquad D_a = e^2. \tag{30}$$

Similarly $\gcd(D_t, D_g) = 1$ and $D_t \geq 1$ give $D_t = \tau^2$, $D_g = g^2$ with $\tau, g > 0$. Since $s \neq 0$, the guard forces $\mathcal{B} \geq 4$, strengthened by its modulo-sixteen class to $\mathcal{B} \geq 12$. It follows that $k, Y \neq 0$, $B_0 > 0$, and

$$Y^2 B_0 > k^2 q^8, \qquad B_0 \leq k^2 q^2, \qquad |Y| > |q|^3.$$

The centered strip from $B_0 > 0$ then excludes $c = 0$.

Define proof-only integers $I = c^2 u$ and $T_{\mathrm{old}} = uT$. By (30),

$$f^2 - \Delta I^2 = 1, \qquad Q = (\Delta I)^2, \qquad V = f^2(c - R + 4IT_{\mathrm{old}}) - c.$$

The rational number $y = e/\delta$ satisfies

$$QV^2 - (Q-1)y^2 = 1. \tag{31}$$

It is integral by adjacent-coefficient descent. If b is its reduced denominator, then $b^2 \mid Q - 1$. After choosing signs, for $|V| > 1$ the unimodular transformation

$$V' = (2Q-1)V - 2(Q-1)y, \qquad y' = (2Q-1)y - 2QV$$

preserves (31) and the reduced denominator, with $0 < |V'| < |V|$. The first coordinate is integral because $(Q-1)y$ is integral; denominator preservation also follows from $\gcd(2Q-1, b) = 1$. For $V > 1$ and $Q > 1$, the norm gives

$$V < y < \frac{(2Q-1)V}{2(Q-1)}.$$

The squared upper bound minus y^2, multiplied by $4(Q-1)^2$, is $V^2 + 4(Q-1) > 0$. Consequently $0 < V' < V$. Descent reaches $|V| = 1$, where $|y| = 1$, forcing $b = 1$. The formal $c = 0$ case would directly give $y = -1$, although it is already excluded here.

The two normalized roots of the original four-conjugate equation are now integers $Lufy$ and τg. Hence Z/S is integral. Since Ψ is integral, $S \mid A_c$. No size or sign condition on Ψ was used. Keep the ten outer coordinates of the new zero and set

$$n_{\mathrm{old}} = A_c/S + 2ufy + \tau g - (2\Delta + 1)UV.$$

This is an integer and makes one old conjugate factor vanish, producing an actual Round 17 zero. Conversely an old zero supplies these same integral roots and divisibility by the retained theorem. Keeping its ten outer coordinates, set

$$n_{\mathrm{new}} = A_c/S + Lufy + \tau g - \Psi.$$

One new factor vanishes. These constructions prove equality of projected zero sets. They need not be mutually inverse on n, since the norm can have several sign branches. For a fixed L, changing only the center is separately a literal bijective integral shear of n. □

6.3 The complete inherited calibration chain

Theorem 6.1 reduces both directions to an actual old zero, not merely a list of necessary Pell properties. Thus its inherited scale, sign, rank and syntactic interfaces remain exactly applicable. For clarity, the retained source proof proceeds as follows.

Carrier divisibility first gives $M \mid H(v - q^3)$, even Y, and

$$H \mid q^{15} K_q^5, \qquad K_q = X(X + 2ac) - 1.$$

The strong and auxiliary pairs, together with $c^2 \mid I$, recover the main Pell root by absolute folding. With $D_t = \tau^2$, the centered strip, $|Y| > |q|^3$ and the authentic packing inequalities

$$|q|(|q|^2 - 1) < R < |q|^4 - 2,$$

the signed rank theorem yields

$$c = \mathrm{sgn}(c)\psi_A(R), \qquad R = 2t + 1, \qquad |H| > |X|(|X| - 3)^t.$$

The powered projection at the actual large base forces $\operatorname{sgn}(c)X = 2^R$. Reduction modulo three fixes $c > 0$ and $X = 2^R$; the old strip fixes $k, Y > 0$. Exact $H \mid K$ restores $Y = (2v_{\mathrm{old}} + 1)q^3$. The uniform half-binomial theorem then restores the positive native interval, and fresh odd strong witnesses produce an actual signed Q41 source zero. Only that restored zero is passed to the retained source34/native/compiler/finite-refutation calibration.

Conversely, canonical source witnesses give $\mathcal{B} > 64$. After choosing the guard pair s, g, the old proof chooses one strong pair I, f with $4sAdc^2 \mid I$, where $D_m = d^2$, and sets

$$u = I/c^2, \qquad t_{\mathrm{strong}} = I/(Adc^2), \qquad r = At_{\mathrm{strong}}^2.$$

That pair is kept fixed. The supported-modulus theorem interpolates T_{old} modulo usr using the same pair, while an intersective root of π modulo sr makes A_c/S integral. There is no reselection of the strong pair after a residue request. The present change replaces only its final integer n by n_{new} above and introduces no new interpolation or choice-order assumption. Full inherited proofs and their source identities are retained separately; they are not promoted here to a newly checked formal-kernel endpoint.

6.4 Exact degrees of the entire output polynomials

The degree claims concern the authentic fixed coefficient tuple, not generic ports, line samples or syntactic upper bounds. The shared register degrees are

Registers	Exact degrees
$a, A;\ \Delta;\ R, k$	$3;\ 6;\ 5$
$\mathcal{B}, D_t;\ P$	$18;\ 56$
$D_m, B, U;\ D_f;\ Q, \delta$	$8, 12, 11;\ 22;\ 28$
$V, C_y;\ S, A_c;\ \Psi, Z$	$37, 102;\ 20, 35;\ 54, 74$
$p_a, p_b;\ 4p_ap_b$	$176, 124;\ 300$

The highest form of k is that of R, since hXY has degree four. Put $W_0 = Sn - A_c$, so $\deg W_0 = 35$. In the free polynomial ring with independent coordinates A, c, r, T, R, assign weights $(3, 1, 1, 1, 5)$ and use the exact definitions $\Delta = A^2 - 1$, $D_m = 1 + \Delta c^2$, $B = rAD_m$, $U = rc^2D_m$, $D_f = 1 + \Delta c^4B$ and $V = D_f(c - R + 4AUT) - c$. Define

$$\begin{aligned} D_{\mathrm{center}} &= \Psi^2 - L^2BD_fV^2, \\ E_{\mathrm{center}} &= \delta\Psi^2 - L^2BD_fC_y \\ &= \delta D_{\mathrm{center}} - L^2BD_f(V^2 - 1). \end{aligned} \tag{32}$$

Exact expansion gives

Center	$\deg D_{\mathrm{center}}$	Highest weighted form	$\deg E_{\mathrm{center}}$
316	90	$-128A^{18}c^{28}r^6T^2$	118
308	86	$-1024A^{19}c^{22}r^5T^2$	114
304	84	$-48A^{16}c^{28}r^6T^2$	112
300	82	$4096A^{16}c^{26}r^6T^2$	110

Every omitted monomial has strictly smaller weight. The following finite identities give a direct route to those expansions. Put

$$\lambda = (8\Delta + 4)U - rc^4, \quad \Psi_0 = \lambda V, \quad h_* = 2U^3T(c^2 - 2), \quad b_* = 16A^3U^2T, \quad e_* = 3rc^6U^2T.$$

Then

$$\begin{aligned} D_{316} &= [8Ur(2c^2 - c^4) + r^2c^8 - 64B]V^2, \\ D_{308} &= D_{316} + 2\Psi_0h_* + h_*^2, \\ D_{304} &= D_{308} + 2(\Psi_0 + h_*)b_* + b_*^2, \\ D_{300} &= 16D_{304} + 8(\Psi_0 + h_* + b_*)e_* + e_*^2. \end{aligned}$$

They follow by squaring the four centers and using $BD_f = B + \Delta A^2U^2$. Substitution of

$$U = r(c^4A^2 + c^2 - c^4)$$

and the displayed formula for V gives the complete finite expansion; the independent checker supplies every exact remainder coefficient. The second term of (32) has weight 108, smaller than each leading $\delta D_{\mathrm{center}}$. Since the highest form of δ is rA^7c^6, the respective highest forms of E_{center} are

$$-128A^{25}c^{34}r^7T^2, \quad -1024A^{26}c^{28}r^6T^2, \quad -48A^{23}c^{34}r^7T^2, \quad 4096A^{23}c^{32}r^7T^2.$$

Now the exact norm cut is

$$p_a + p_b - p_c = \delta W_0^2 + 2\delta S\Psi W_0 + \delta S^2 P + S^2 E_{\mathrm{center}}. \tag{33}$$

The first three terms have degrees at most 98, 137 and 124. The last has exact degree 158, 154, 152 or 150, respectively, with a unique highest monomial after the actual substitution $A = a + 2$. Squaring therefore gives degrees 316, 308, 304 and 300. In the first three cases, $4p_ap_b$ has smaller degree 300. In the last case its highest form has T-degree two, whereas the highest monomial of the squared inner term has T-degree four. That monomial cannot cancel.

For explicit coefficients, put $q_0 = 6\beta j$ and $a_0 = 12q_0wv$. Then $S_0 = 128sra_0q_0^{15}$. Surviving highest monomials have the following exponents, with all omitted variable exponents zero:

Degree	Exponents in (j, w, v, s, r, c, T)	Positive coefficient
316	$(114, 54, 54, 4, 18, 68, 4)$	$2^{264}3^{168}\beta^{114}$
308	$(116, 56, 56, 4, 16, 56, 4)$	$2^{276}3^{172}\beta^{116}$
304	$(110, 50, 50, 4, 18, 68, 4)$	$2^{246}3^{162}\beta^{110}$
300	$(110, 50, 50, 4, 18, 64, 4)$	$2^{262}3^{160}\beta^{110}$

Each exponent sum is the stated degree and each coefficient is nonzero for the actual β. For degree 300, the uncancelled T^4 summand before expanding a_0, q_0 is

$$2^{52}s^4r^{18}c^{64}T^4a_0^{50}q_0^{60}.$$

This establishes exact attained degrees. It is not a lower bound for all possible eleven-variable constructions.

6.5 Live costs and finite checks

The actual graphs have the following operation counts:

Degree	Additions	Subtractions	Multiplications	Total
316	27	21	85	133
308	28	22	90	140
304	30	22	92	144
300	31	22	95	148

A square or multiplication by a fixed numeral costs one binary gate; register reuse is free. All eleven variables and six ports are live. The independently implemented checker imports neither the producer nor its graph class. It binds all named registers and live gates in free polynomial rings, verifies the four-conjugate identity, center remainders, degree cuts and actual leading coefficients, authenticates and reconstructs all six numerical payloads, checks source pins and liveness, and rejects altered constants, coordinates, degree claims and selected semantic gates. These are finite algebraic checks, distinct from the retained unbounded Pell, interpolation, intersectivity and native-refutation theorems.

7 Comparison with the six incoming packages

7.1 Immutable intake and the comparison boundary

The inspected incoming snapshot is ProveIt commit

`e0c98c629bf10d30b695884f6b1d2ab361bdf513`.

It was authored at 20:29:46 UTC on 25 October 2026, 241 reachable commits after the preceding comparison pin `1c5367452a0bd4dd7e9e4bceb1508b62dbd2ff61`. Exact directory-object comparison found no new or changed numbered construction directory through source133 in the established Diophantine-consistency collection; its README had changed. This conclusion uses complete directory listings, not a truncated repository-wide file list.

All six incoming archives were added at the pinned head. They were obtained from immutable URLs, authenticated against their listed Git blob SHA-1 values, checked for escaping archive paths, and extracted. The retained `remote_manifest.json` records exact names and Git identities. The intake synopsis records which statements are incoming reports and which new objects were independently rebuilt. Arrival in `docs/incoming` is not acceptance of the mathematics or integration into active code.

7.2 Natural quartics

All rows in this table concern degree-four equations over independent nonnegative integer coordinates. The incoming metrics below distinguish finite algebra from full mathematical acceptance. A separate implementation recollected every coefficient of nine incoming direct endpoints: Certificate reductions 314 and 233, Local certificates 306, Semantic signs 286, Odd helpers 307, 308 and 310, and Graph digits 226 and 227. Each expansion, dimension, degree, support and height agreed with its literal artifact. No incoming producer was executed for that check; no whole-package theorem acceptance is implied. The other entries are reported incoming values or the authenticated prior release.

Incoming endpoint	Variables	Rows	Terms	Height
Certificate reductions	233	182	7313	86543666
Round 17 principal	229	179	9184	83954634
Round 17 sparse	230	180	6858	83954472
Semantic signs	235	184	22185	90911633
Semantic signs sparse	236	185	15645	88233865
Graph digits principal	226	175	10314	87839583
Graph digits sparse	227	176	7905	87839583

The incoming Certificate reductions package contributes a descending factorial fixed point and different polynomial Pell comparison center. The Graph digits package contributes shared low-digit budgets. The present natural construction composes that pooling idea with the Round 17 aligned self-sized radix rather than treating the incoming result as merely a numerical benchmark. The exact final comparisons use the new measured endpoints in the opening table; in particular, a smaller variable count alone is not called domination when support or height increases.

7.3 Unrestricted quartics

Every coordinate in the next table is an unrestricted integer; no natural inequality is implicit in a variable name.

Incoming endpoint	Variables	Rows	Terms	Height
Certificate reductions	314	204	5721	128446
Round 17 principal	282	192	5491	128446
Round 17 grouped	287	197	4313	128446
Round 17 sparse	309	219	3922	128446
Local certificates	306	206	6259	128446
Odd helpers minimum	307	211	22532	394412
Odd helpers balanced	308	212	5364	195962
Odd helpers low height	310	214	5612	128446
Semantic signs principal	286	196	6607	128446
Semantic signs generated	319	207	5779	128446

The Local certificates and Semantic signs packages recover orientation and sign information from existing Pell data. Round 17 already retained a complete orientation-first proof for its smaller direct sources. The new signed reduction uses a different resource: canonical low digits occupy only a small part of their denominator interval, so several interval certificates can be replaced by one spherical bound. The proof must show that a possibly negative purported remainder is the canonical nonnegative tail; dropping separate signs without that argument would not be valid.

The Odd helpers package also supplies separate 13- and 14-variable high-degree equations. Their different dimensions and degrees are not mixed with the quartic comparison above.

7.4 Height one with a source controlled comparison

The incoming height-one equations all have exact degree four, unrestricted integer coordinates and constant coefficient one. Their reported variable/support pairs include

Source package	(variables, terms)
Round 17	$(793, 51982)$, $(806, 23719)$, $(994, 11744)$
Local certificates	$(985, 95609)$, $(1122, 25922)$
Graph digits	$(951, 56916)$, $(984, 30076)$, $(1050, 21766)$.

The Local and Graph digits matched-source controls use the older 442-variable signed source and report $(1252, 27738)$ and $(1137, 23105)$, respectively. These controls isolate a compiler change from the separate effect of choosing a smaller arithmetic source.

A new matched-source replay fixes exactly the same refactored 287-variable source, 48-hub setting and seed zero as the Round 17 806-variable, 23719-term graph. The new compiler gives 603 variables and 12289 terms on that same source. This isolates the combined normalization, row-sign and incidence-network improvement from any change to the arithmetic source. The complete receipt and replay are retained in the height-one comparison component.

The present graph mechanism directly pairs opposite signed occurrences of the same logical monomial in distinct rows. This changes the incidence network before the existing alias and equality compression. A comparison on the same fixed source therefore establishes a genuine compiler gain; additional improvement from a new spherical source is reported separately. Row sign choices change the source presentation but not the simultaneous zero set. They are certified explicitly rather than attributed to an unproved optimal graph heuristic.

7.5 Small variable objectives and domain separation

The incoming eleven-integer points include degree/cost pairs

$$(924, 107), \quad (840, 108), \quad (648, 115),$$

from Semantic signs and Odd helpers, as well as the Round 17 family

$$(328, 130), \quad (340, 129), \quad (352, 125), \quad (398, 121).$$

Each cost counts live binary additions, subtractions and multiplications under the stated convention. Low-degree higher-cost points do not erase the lower-cost high-degree alternatives. The new scalar shear is evaluated with the exact authentic fixed tuple, not merely as a generic symbolic change of coordinates.

The positive ten-variable branches are different objectives. Incoming Certificate reductions reports cost/degree 78/630 and 79/468; Semantic signs reports 83/436, 81/560, and 80/750; Graph digits reports 87/323 and 88/318; Odd helpers reports 81/750 and 82/560. These use strictly positive witnesses and a different coefficient tuple. They are relevant research comparisons, but they are not evidence that an unrestricted eleven-variable construction has ten variables, nor are their fixed coefficient ports interchangeable with source34.

7.6 Attribution and limits

The retained ingredients include the numerical C222 calibration, ordinary Pell and indexed-Pell theory, reciprocal factorial CRT, square-radius factorial recovery, semantic anchors, cofinal representatives, self-sized radices, signed orientation, native coefficient compilation, signed rank, interpolation, incidence energies, alias contraction, and quartic equality energies. The new sections identify the additional transformations rather than renaming these inherited mechanisms.

The comparison universe is the inspected pinned collection and six incoming archives, not the whole literature. No worldwide priority claim, global dimension minimum, arithmetic-circuit lower bound, or external human peer review is implied. The new independent audits are separate reasoning and implementations within this AI-assisted workflow.

8 Reproduction and evidence boundaries

8.1 Portable entry points and normative artifacts

From the extracted release root, the integrated computational entry point is

```
python3 -B verify_release.py
```

It checks the release inventory and invokes the explicit component plan, including normal and optimized interpreter runs and deterministic rebuilds. The mathematical checks require Python and its standard library, not a network connection, repository checkout, external computer-algebra system or theorem-prover installation. Rebuilds and any receipt-writing inherited checks run in disposable copies so they cannot alter the sealed reference files. The final integrated replay receipt records the tests actually run; a command listed here is not by itself evidence that it passed.

The direct equations are the complete residual arrays and adjacent expanded coefficient arrays. The selected signed paths are the dimension, grouped and sparse `centered_merged_odd` variants and the sparse `centered_odd` four-pool variant. The natural catalog identifies `natural221`, `natural222`, `natural223` and `natural224_sparse`. The height-one catalog identifies `compact571`, `balanced593`, `sparse646` and `sparser742`. The scalar circuits are `free_whole_center11_c316.json`, `free_whole_center11_c308.json`, `free_whole_center11_c304.json` and `free_whole_center11_c300.json`. Each component's catalog and binding file give exact paths and hashes; these are not unspecified polynomial families.

The editable report has a separate build command, from the release root,

```
sh report/build_report.sh
```

A normal TeX installation is needed only for that step. The builder uses a fresh temporary directory, creates a local format when the installed pdfLaTeX format is absent, runs three passes, and rejects overfull boxes and unresolved references. Mathematical verification does not require TeX.

8.2 Recorded component checks

The following results were obtained before report finalization.

- The signed independent checker reconstructed all twelve emitted variants in named coordinates, checked every retained and changed row, and collected every coefficient. Normal and optimized runs passed; 30 deliberate coefficient corruptions were rejected. A separate audit rejected 60 raw-transformation mutations and rechecked the unbounded spherical, merged and odd-square arguments.
- The natural checker verified all four endpoint expansions and nonnegative restoration maps, including the inherited 618-gate, 397-restoration affine export to the 334-coordinate raw source. It checked 37581 collected endpoint coefficients, rejected 21 negative controls, and reproduced 20 mathematical files byte for byte. Normal and optimized isolated component runs passed. A separate raw-transformation audit rejected 15 changed-row controls.
- The height-one checker verified every occurrence boundary, source pullback, graph and tree invariant, equality class, literal coefficient, and symbolic lift at every vertex. All four profiles passed normal and optimized replay, 13 corruption classes per profile were rejected, and selected source transformations and outputs regenerated byte for byte. The same-source comparison from 806 variables and 23719 terms to 603 variables and 12289 terms also passed its separate replay.
- The scalar checker authenticated all six coefficient payloads and reconstructed their complete finite recipe. It checked the universal norm, center identities, exact polynomial remainders, leading-degree cuts, attained actual coefficients, circuit counts, liveness and source pins. Normal and optimized runs passed, rejecting 1130 gate mutations in each mode. All four circuit files and the metric catalog reproduced exactly.
- A separate incoming-artifact implementation recollected all coefficients of the nine direct comparison equations identified in Section 7; each agreed with its supplied expansion. No incoming producer was needed for this comparison check.

These are finite algebra, authentication and regression checks. The unbounded mathematical audits are ordinary arguments within this AI-assisted workflow, not outside human review or a proof-assistant build.

8.3 Inherited source closure

The archive retains the complete authenticated direct and architecture trees under `prior_direct_closure`. They include the numerical calculus, Pell and CRT arithmetic, indexed-Pell criterion, factorial fixed point, constructor bounds, orientation-first argument and strict quotient proof, together with the named logical interfaces. Their historical READMEs describe their original package, not a new execution claim in the present release. Active bindings identify the exact inputs used by the new components.

The scalar component separately retains the full source34 numerical recipe and native/compiler interface, signed-quadratic and rank arguments, carrier and saturation proofs, supported-modulus interpolation, and the Round 17 linear-cofactor theorem. The mathematical handoff is explained in Section 6: a new zero is converted to an actual old zero before any inherited refutation theorem is invoked. The height-one theorem is an unconditional transformation of a certified integral quadratic system; its consistency meaning comes only through its supplied signed source.

8.4 What successful replay cannot establish

Authentication establishes file identity, not mathematical truth. Coefficient replay establishes exact algebra and literal artifact binding, not an unbounded classification theorem. The ordinary

proofs establish witness implications, conditional on their identified inherited inputs; no finite search for zeros is used. The logical calibration concerns syntactic refutation, not a claim that ZFC is consistent or inconsistent. Retaining Lean interface files is not a substitute for rebuilding a transitive kernel-checked development. The manifest and visual rendering receipts similarly certify the delivered files and their presentation, not a new logical endpoint.

9 Further questions

9.1 Joint optimization of arithmetic and graph sources

A source with fewer coordinates need not yield the smallest height-one polynomial: its monomial multiplicities, opposite-sign opportunities and alias incidences also matter. Can arithmetic refactorization, row-sign balancing and occurrence matching be chosen jointly with a useful certificate of optimality for a fixed finite source? A rigorous lower bound for one constrained compiler class would be meaningful; it would still not be a lower bound for all Diophantine equations.

9.2 Sharper pooled digit regions

The present bounds exploit generous known radix margins. One may ask whether an ellipsoidal form with certified integral weights permits further pooling across unequal denominators without increasing degree or adding more coordinates than it removes. Soundness must still identify the actual integer remainder, including negative candidates, before factorial or range decoding. A numerical search for small examples is not enough.

9.3 Less expensive exact polynomial shears

An integral coordinate shear can lower degree while increasing graph cost. Can one share its polynomial-part calculation with the existing carrier registers to retain the degree gain using fewer live binary operations? The relevant objective is the full fixed-coefficient output graph, including cancellations in the highest homogeneous component, not the apparent degree of an intermediate expression.

9.4 Formalizing the transformation boundary

The new finite row transformations, monic witness restorations and network energy identities are natural targets for machine-checked formalization. A useful next milestone is a checked theorem connecting the emitted arrays to the already identified source relation, followed by a transitive build of the calibration boundary. Merely checking a supplied hash manifest or including a named Lean declaration does not accomplish that milestone.